Lesson 08.2: Causal Inference

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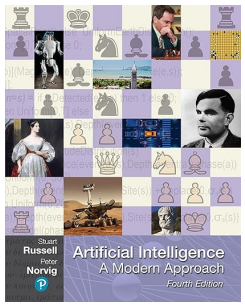
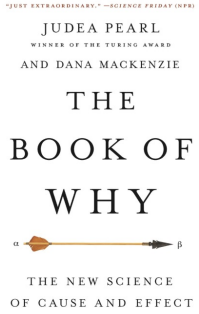
References:

- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017





- ***Causal Networks***
  - Causal DAGs
  - Structural Causal Model
  - Variables
  - Type of Variables in Causal AI
  - Types of Paths in Causal AI

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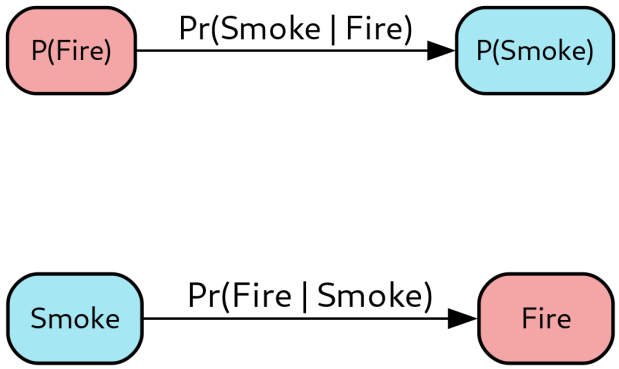
(Non-Causal) Bayesian Networks

- **Bayesian networks** represent a joint distribution function
  - The direction of the arrow represent *conditional dependence* (not causality)
  - $A \rightarrow B$ requires to estimate $\Pr(A|B)$
- **Many possible Bayesian networks** with same nodes, different edges to explain the same phenomenon



- **Example**
  - A Bayesian network with *Fire* and *Smoke*, which are dependent
  - $Fire \rightarrow Smoke$
    - Need $\Pr(Fire)$ and $\Pr(Smoke|Fire)$ to compute $\Pr(Fire, Smoke)$
  - $Smoke \rightarrow Fire$
    - Need $\Pr(Smoke)$ and $\Pr(Fire|Smoke)$
- **Different Bayesian networks**:
  - Are equivalent and convey the same information
  - Have different difficulties to be estimated

Causal (Bayesian) Networks

- **Causal networks are Bayesian networks with only causal edges**
  - Use judgment based on nature instead of just statistics
  - E.g., you need to go from
    - *"Are random variables Smoke and Fire correlated?"* to
    - *"What causes what, Smoke or Fire?"*
- **"Dependency in nature"** is like assignment in programming
  - E.g., nature assigns *Smoke* based on *Fire*:
    - ✅ $Smoke := f(Fire)$
    - ❌ $Fire := f(Smoke)$
- **Structural equations** describe "assignment mechanism" in causal graphs

$$X_i := f(X_j) \iff X_j \to X_i$$

Causal DAG

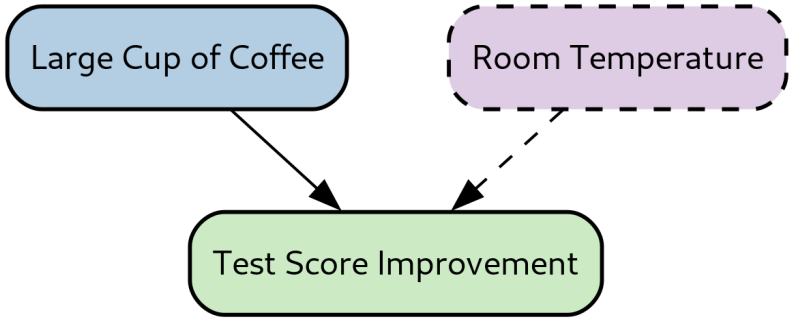
- **Causal DAG**
  - *Directed*: Arrows show cause $\rightarrow$ effect
  - *Acyclic*: No feedback loops
    - Causal relationships assume temporal order: cause before effect
    - A cycle implies a variable is both cause and effect of itself
- **Benefits**
  - DAGs makes explicit *causal* links
  - Support explainable AI models
  - Stability in conditional probability estimation
  - Reason about interventions and counterfactuals
- **Limitations**
  - Requires domain knowledge for structure
  - Assumes all relevant variables included (no hidden confounders)

Causal Edges are Stable

- **Causal edges reflect stable relationship**
  - *Mechanistic stability*
    - Causal relationships show system function, not just behavior in one dataset
    - E.g., "*Temperature* $\rightarrow$ *ice melting rate*" holds true in Alaska and Arizona
  - *Invariance under interventions*
    - If X causes Y, intervening on X affects Y consistently, despite confounders or context changes
  - *Easier estimation through causal modeling*
    - Identifying causal direction focuses estimation on effect size (e.g., regression of Y on X under intervention)
- **Example**: study *Exercise* $\rightarrow$ *Health*:
  - Correlation may differ in young or elderly populations
  - Causal effect remains stable, as physiological mechanism doesn't change

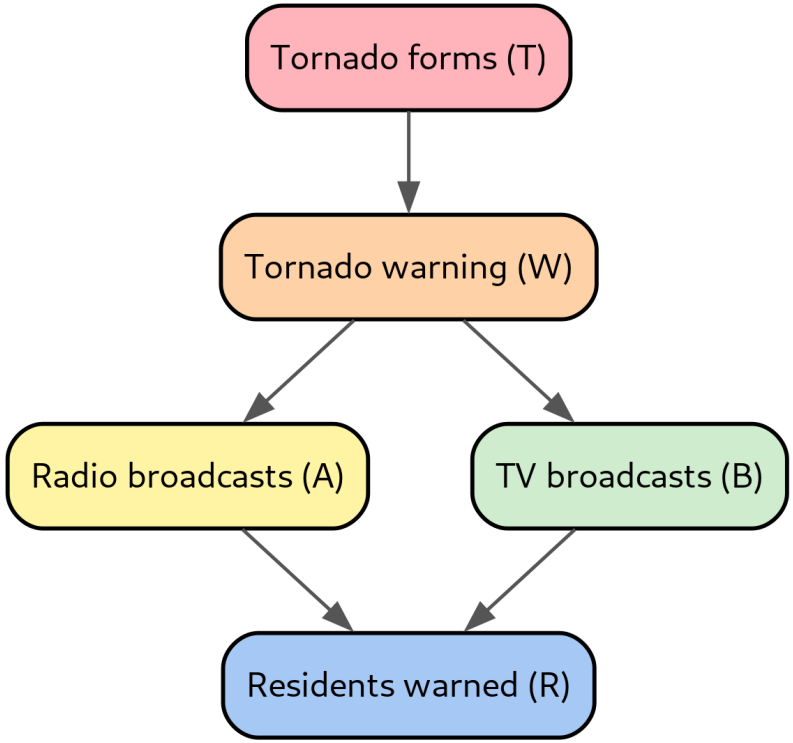
Causal DAG: Example

- **Explanatory variables**
  - You can manipulate or observe when changes are applied
  - E.g., *"does a large cup of coffee before an exam help with a test?"*
- **Outcome variables**
  - Result of the action
  - E.g., *"by how much did the score test improve?"*
- **Unobserved variables**
  - Not seen or more difficult to account
  - E.g., *"temperature of the room makes students sleepy and less alert"*



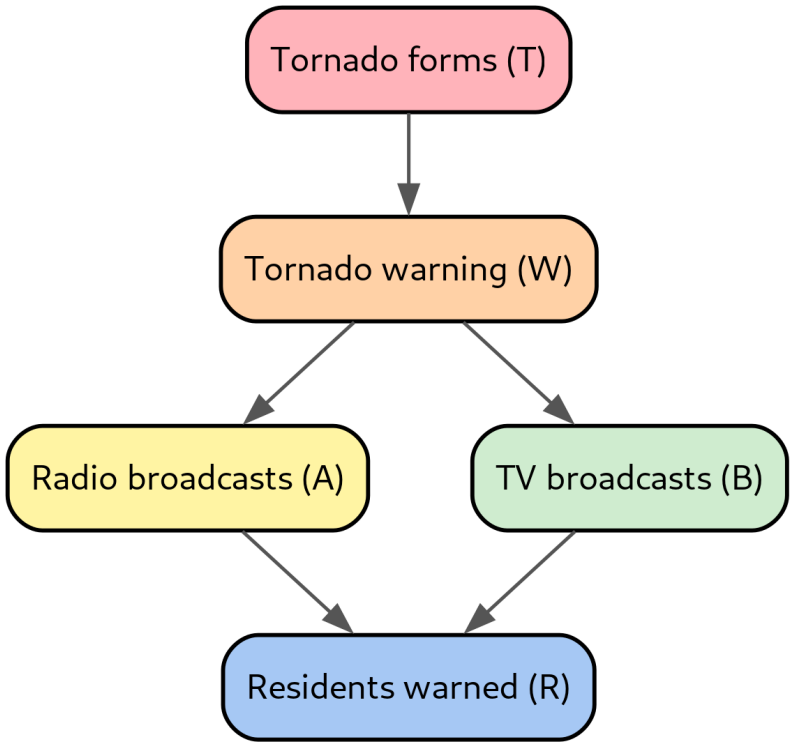
Example: Tornado Warning

- **Example**: Severe weather alert to the public
- **Causal diagram**:
  - T: Tornado forms
  - W: National Weather Service issues tornado warning
  - A, B: Radio and TV broadcast emergency
  - R: Residents warned
- **Assumptions**
  - Channels broadcast only on official order
  - Delivery systems never fail
  - If either channel activates, residents warned



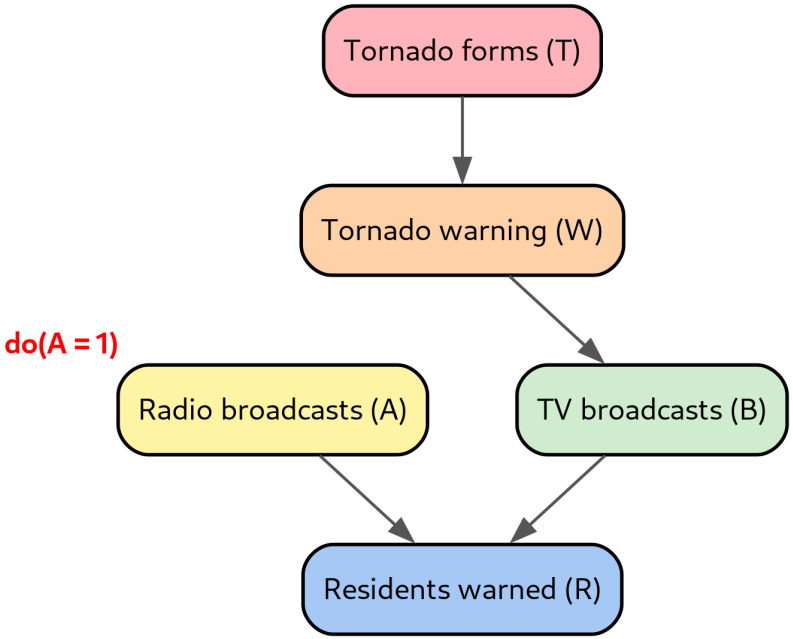
Tornado Warning: Level 1 (Association)

- **Use causal graph to understand how one fact informs another**
  - Use predicate logic to answer
- **If residents are warned, was a warning issued?**
  - Does $R \implies W$?
  - Yes
- **If radio A broadcast, did TV B also broadcast?**
  - Yes
  - True even though A doesn't cause B
  - Correlated through common cause W (confounder)



Tornado Warning: Level 2 (Intervention)

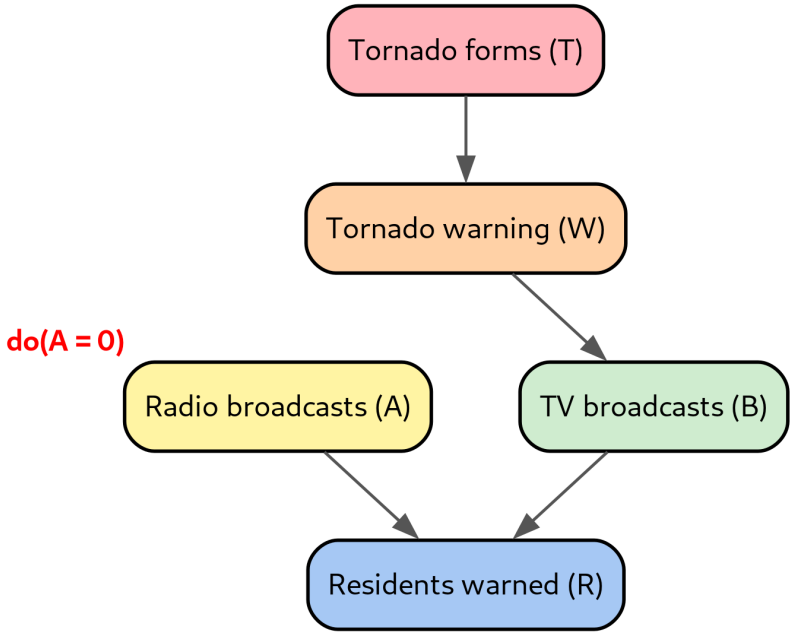
- **If you make radio broadcast, will residents be warned?**
  - Yes
  - Breaks "rules of nature" as A broadcasts only if W is issued
    - Like removing edge $W \to A$ in graph
    - Set A to *true*: $do(A = 1)$
  - Expect B did *not* broadcast
    - Forcing A shouldn't affect B (no $A \to B$ edge)



- **Key difference between "seeing" and "doing"**
  - If *see* A broadcast, conclude B broadcast too
  - If *make* A broadcast, expect B didn't
- **Observational data wouldn't reveal causal structure**
  - Either all variables T, W, A, B, R are true or all are false
  - Correlations alone are uninformative

Tornado Warning: Level 3 (Counterfactual)

- **Assume that residents are warned $R = 1$**
  - You know A and B broadcast because
  - ... a warning W was issued
  - ... after a tornado formed T
- **Would residents be warned if A had not broadcast?**
  - Remove the edge into A
  - Set $do(A = 0)$
  - TV B still broadcasts (due to W), and $R = (A \vee B)$ remains true
  - Yes, $R = 1$



- Causal Networks
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  - ***Structural Causal Model***
  - Variables
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Structural Causal Model

- A **Structural Causal Model** (SCM) translates a causal DAG into mathematical equations to define how variables interact
- **Structure of SCMs**
  - *Variables* $X_1, X_2, ..., X_n$ represent quantities in the system
  - *Equations* model each variable as a function of its direct causes
  - Formally, X_i is modeled as:

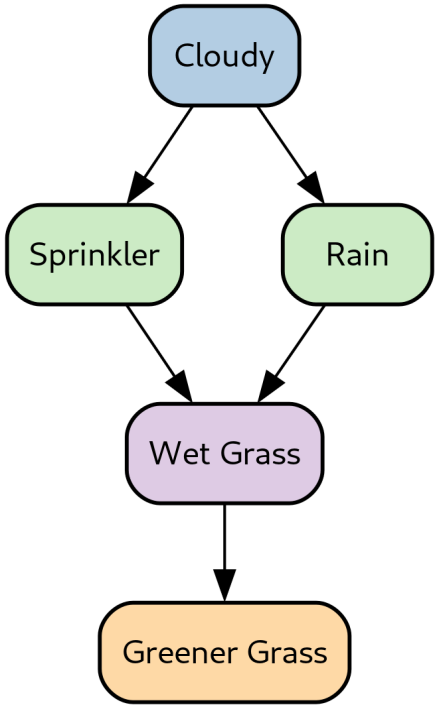
$$X_i = f_i(Parents(X_i), \varepsilon_i)$$

    where:
    * $Parents(X_i)$ are direct causes of X_i
    * ε_i is an exogenous (external, unobserved) noise term
- **Properties**
  - Same properties of causal networks
    * Explain causal relationships between variables
    * Provide a foundation for causal reasoning and simulation
    * ...
  - Quantify effect
- Used in econometrics and genetics for a long time (even before theory of

Structural Causal Model: Sprinkler Example

- **Structural equations** for this causal DAG:

$$\begin{cases} C := f_C(\varepsilon_C) \\ R := f_R(C, \varepsilon_R) \\ S := f_S(C, \varepsilon_S) \\ W := f_W(R, S, \varepsilon_W) \\ G := f_G(W, \varepsilon_G) \end{cases}$$



- **Unmodeled variables** ε_x represent error terms
  - E.g., ε_W is another source of wetness (e.g., *MorningDew*) besides *Sprinkler* and *Rain*
  - Assume unmodeled variables are exogenous, independent, with a certain distribution (prior)
- Express **joint distribution** of all variables as a product of conditional distributions using causal DAG topology:

$$\Pr(C, R, S, W, G) = \Pr(W|R, S)\Pr(G|W)\Pr(S|C)\Pr(R|C)\Pr(C)$$

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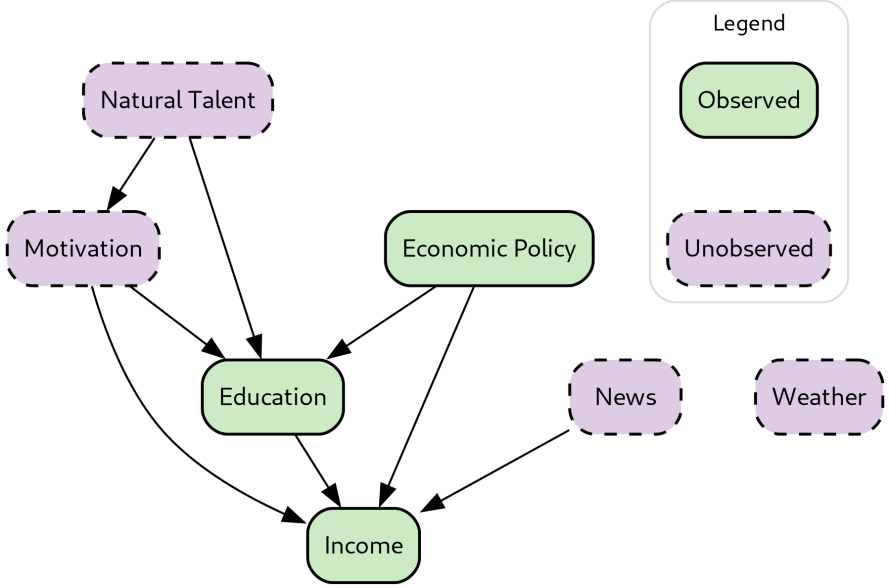
Observed Vs. Unobserved Variables

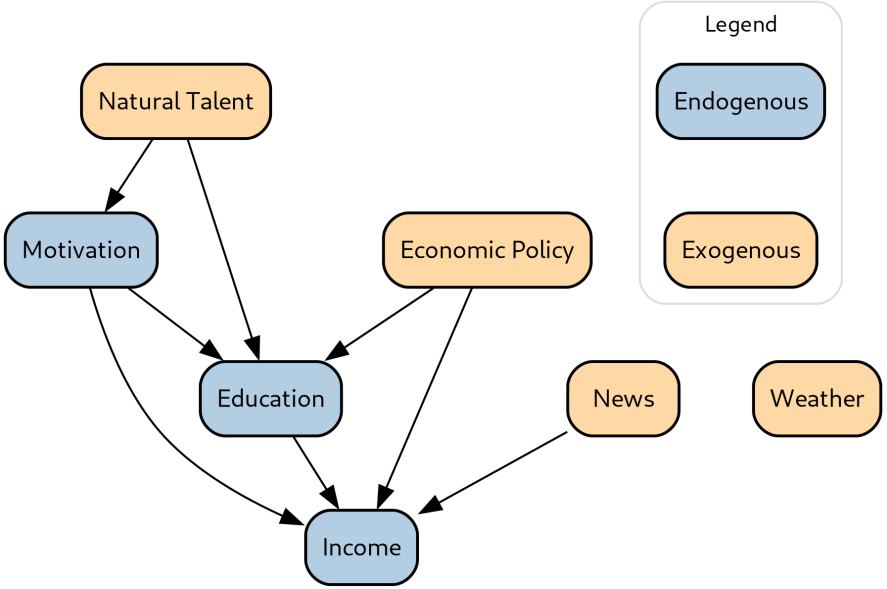
- **Observed variables**
  - Aka "measurable" or "visible"
  - Variables directly measured or collected in a dataset
  - E.g.,
    - Education
    - Income



- **Unobserved variables**
  - Aka "latent" or "hidden"
  - Exist but not measured or included in data
  - E.g.,
    - Natural talent
    - Motivation
  - Ignoring unobserved variables leads to incorrect conclusions
    - E.g., $IceCreamSales \leftarrow Temperature \rightarrow DrowningRates$

Endogenous Vs. Exogenous Variables

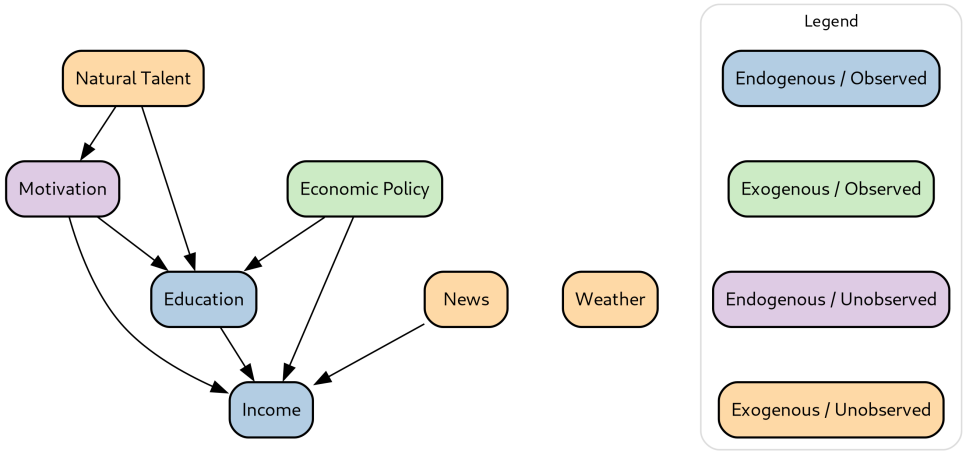
- **Endogenous variables**
  - Values determined *within* the model
    - Dependent on other variables in the system
  - Represent system's internal behavior and outcomes
  - E.g.,
    - Motivation
    - Income



- **Exogenous variables**
  - Originate *outside* the system being modeled
    - Not caused by other variables in the model
  - Represent background conditions or external shocks
  - E.g.,
    - Natural talent
    - Economic policy
    - Weather

Endo / Exogenous, Observed / Unobserved Vars

- **Typically**
  - *Exogenous / unobserved variables*: capture randomness or unknown external factors
  - *Exogenous / observed variables*: potential intervention factors
  - *Endogenous / observed variables*: focus for prediction and intervention



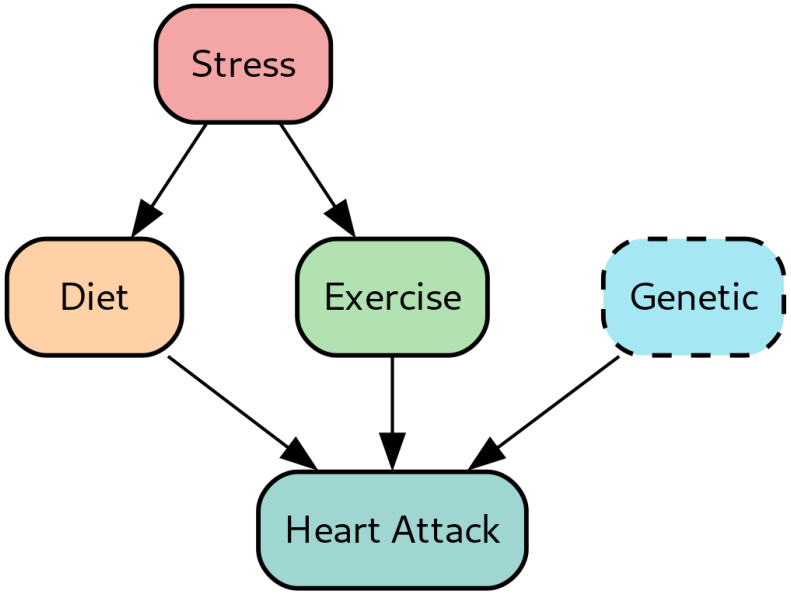
Variable Type	Observability	Example
Endogenous	Observed	Income
Exogenous	Observed	Education
Endogenous	Unobserved	Motivation
Exogenous	Unobserved	Natural Talent

Building a Causal DAG

- **Causal models** visually represent complex environments and relationships
  - Nodes are like "nouns":
    - E.g., "price", "sales", "revenue", "birth weight", "gestation period"
    - Variables can be endogenous/exogenous and observed/unobserved
  - Relationships between variables are "verbs":
    - Parents, children (direct relationships)
    - Descendants, ancestors (along the path)
    - Neighbors
- **Modeling as a Communication Tool**:
  - Shared language bridges gaps between technical and non-technical team members
- **Iterative Refinement**:
  - Continuously update models with new variables and insights

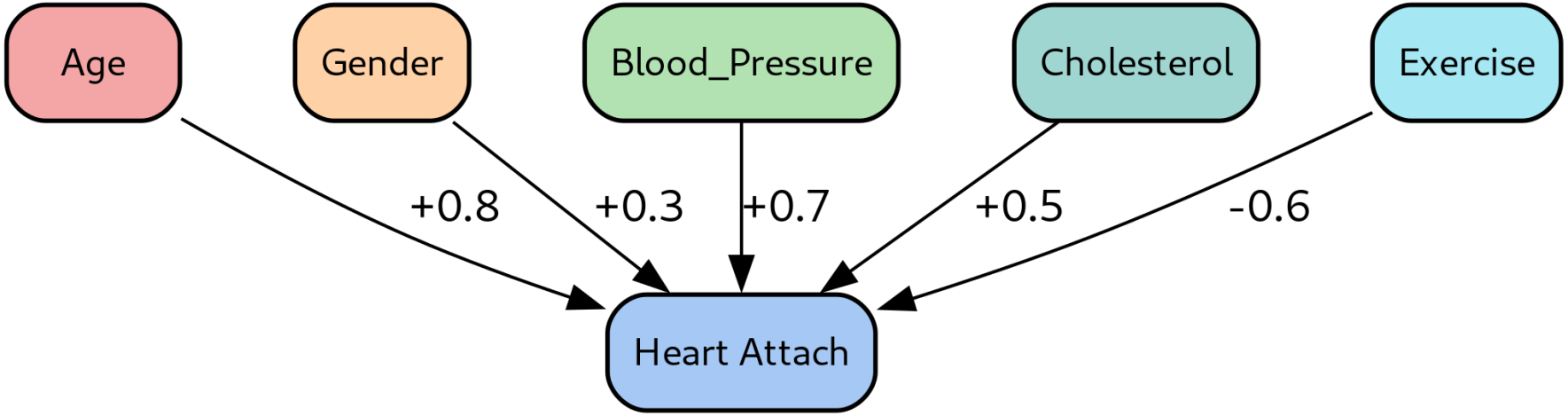
Heart Attack: Example

- Research question: *What's the relationship between stress and heart attacks*?
- **Build a causal DAG**
  - *Stress* is the treatment
  - *Heart attack* is the outcome
  - Stress is *not a direct cause* of heart attack
    - E.g., a stressed person tend to have poor eating habits and tends not to exercise
  - *Genetics* is unobserved



Weights

- Assign weights to paths to represent causal strength
- Sign indicates direction



- **How to estimate sign and weight**
  - Estimate using correlation
  - Use priors and then estimate using Bayesian approach

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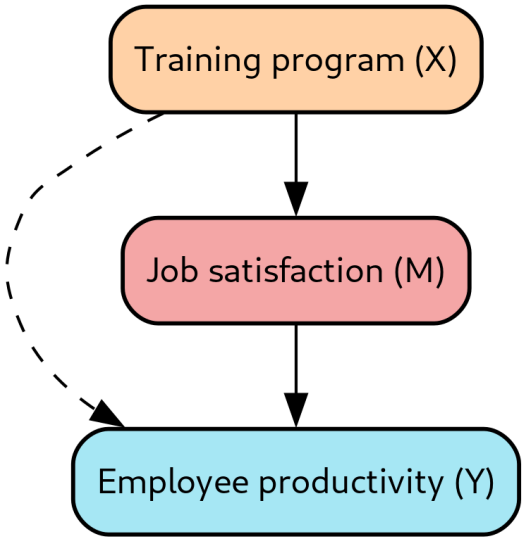
Mediator Variable

- A **mediator variable** M
  - Is an intermediate variable that *transmits* the causal effect from X (treatment) to Y (outcome)
  - Lies **on the causal path** between X and Y
  - Captures the **mechanism or process** through which X influences Y

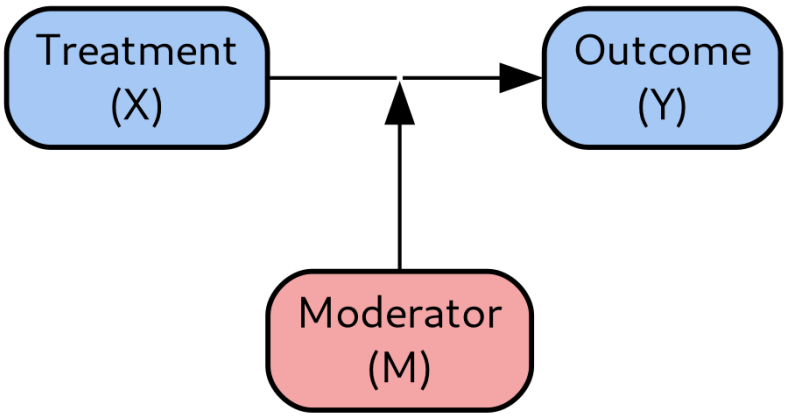
Mediator Variable: Example

- **Research question**: *"Does a training program increase employee productivity?"*
- Causal effect may be indirect, through a **mediator**
  - Training might not immediately boost productivity
  - Could enhance job satisfaction, raising productivity
- **Causal interpretation**
  - X: Training Program (cause)
  - M: Job Satisfaction (mediator)
  - Y: Employee Productivity (effect)
  - Path: $X \to M \to Y$
- **Direct vs. Indirect effects**
  - *Indirect effect*: X affects Y through M
  - *Direct effect*: X affects Y not through M
  - Controlling for M separates effects, clarifying training impact



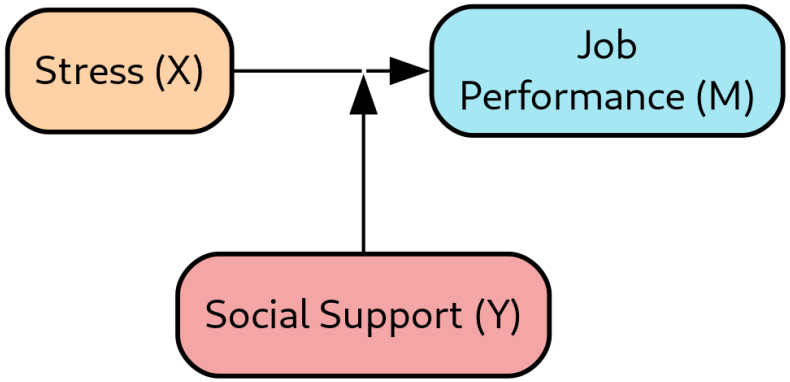
Moderator Variable

- A **moderator variable** M
  - Changes the *strength* or *direction* of the relationship between an independent variable (X) and a dependent variable (Y)
  - Is not part of the causal chain but conditions the relationship



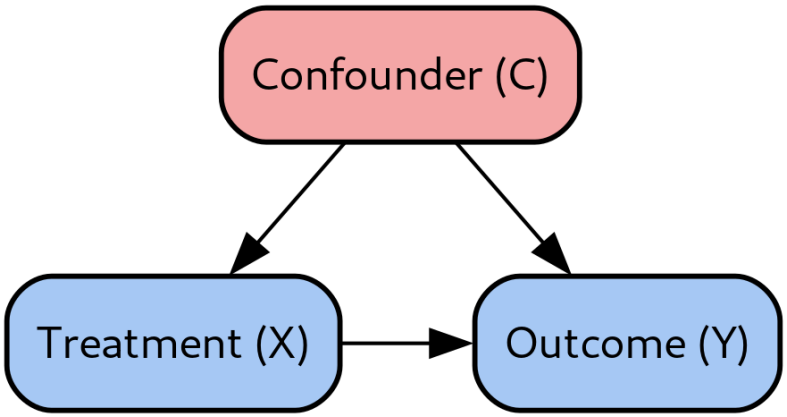
Moderator Variable: Example

- **Research question**: *"Study relationship between stress and job performance"*
- **Social support** M as a moderator
  - High social support weakens stress's negative effect on performance
  - Low social support strengthens stress's negative effect on performance



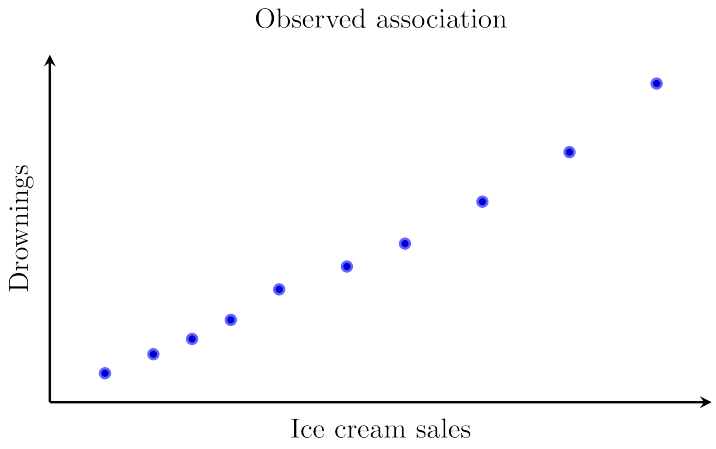
Confounder Variable

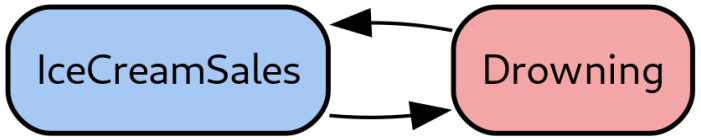
- A **confounder** C
  - Affects both treatment (cause) and outcome (effect)
  - Creates misleading association, if not controlled



Confounder Variable: Example

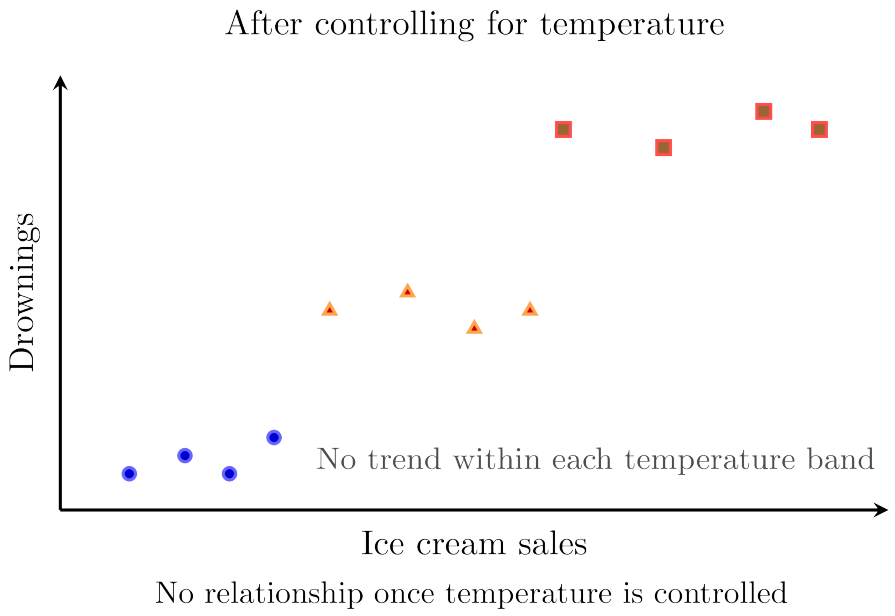
- *IceCreamSales* and *Drowning* **move together**
  - Correlation-based model claims association
  - Is it true?
  - You can always find an explanation (e.g., from ChatGPT)
    - Eating ice cream may distract children or guardians:
    - Cold food shock reflex causes hyperventilation in water
    - Sugar spike $\rightarrow$ hyperactivity near pools
- **How to use this relationship?**
  - Ban ice cream to prevent drowning?
  - Ice cream maker increase drowning to boost sales?

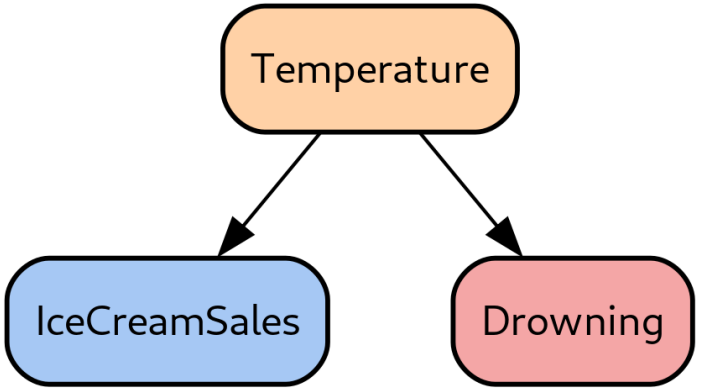




Confounder Variable: Example

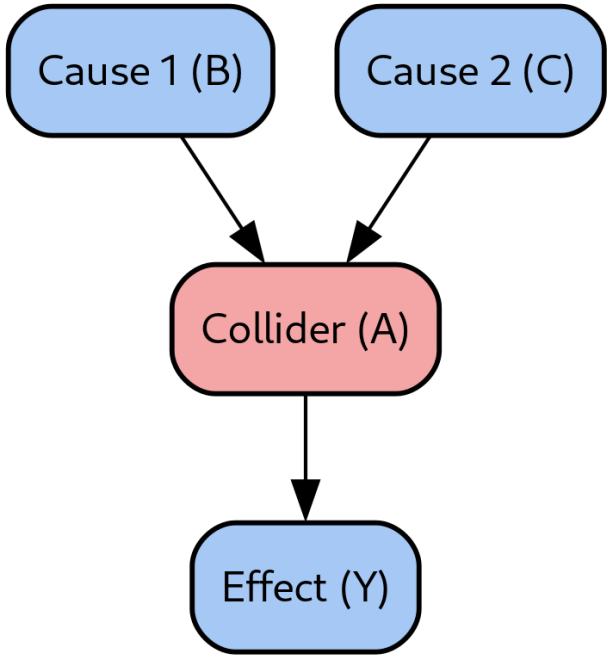
- In reality, **no cause-effect** between *IceCreamSales* and *Drowning*
  - *Temperature* is a confounder
- In fact, when control for temperature (in regression or intervention), association disappears





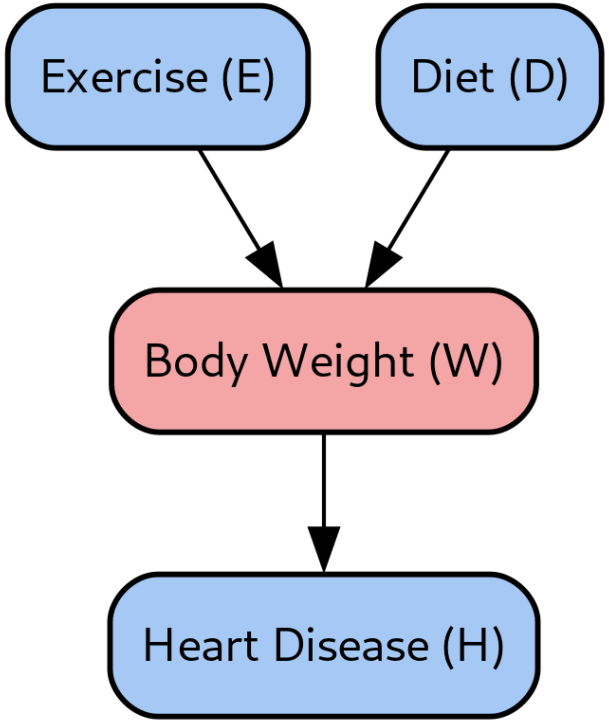
Collider

- A **collider** A
  - Is a variable influenced by multiple variables B, C
  - Complicates understanding relationships between variables B, C and those it influences, Y



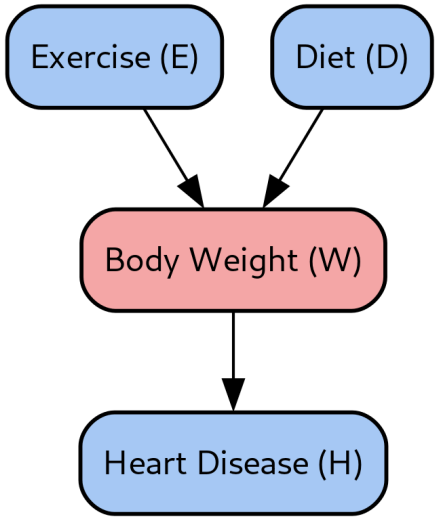
Collider: Examples

- Study the relationship between *Exercise* and *HeartDisease*
  - *Diet* and *Exercise* influence *BodyWeight*
  - *BodyWeight* influences *HeartDisease*
  - *BodyWeight* is a collider



Collider Bias

- Aka "Berkson's paradox"
- **Conditioning on a collider** can introduce a spurious association between its parents by *"opening a path that is blocked"*
- **Example**
  - *Diet* (D)
  - *Exercise* (E)
  - *BodyWeight* (W)
  - *HeartDisease* (D)
- **Without conditioning on W**
  - E and D are independent
    - E.g., knowing exercise level E doesn't inform about diet D, and vice versa
  - Collider W blocks association between E and D
- **After conditioning on W**
  - E.g., individuals with specific body weight
  - Introduce dependency between E and D
  - With W fixed, changes in E balanced by changes in D, inducing spurious correlation between E and D



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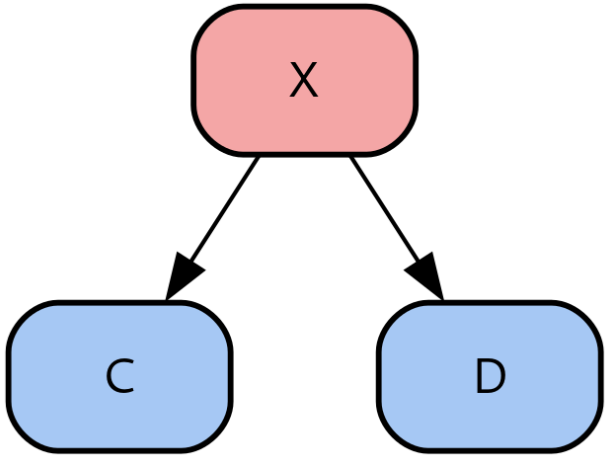
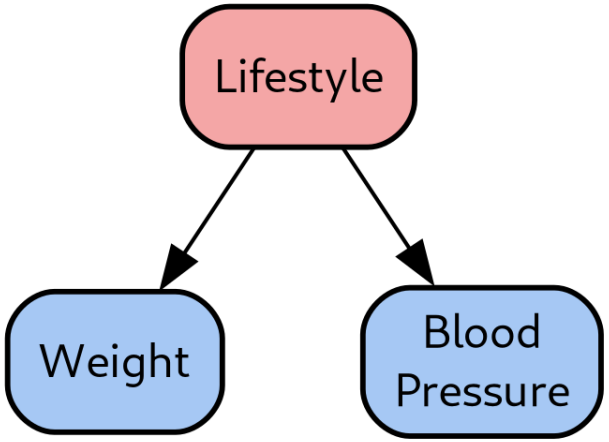
Fork Structure

- A **fork** $D \leftarrow X \rightarrow C$ occurs when a single variable causally influences two or more variables
  - X is a **confounder** (common cause) of C and D
  - Forks induce statistical dependence between C and D even if C and D are not causally linked
- **Conditioning** on X *blocks the path* and removes spurious correlation

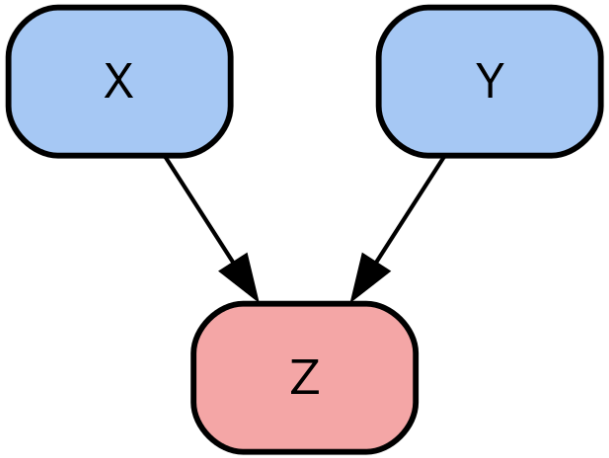
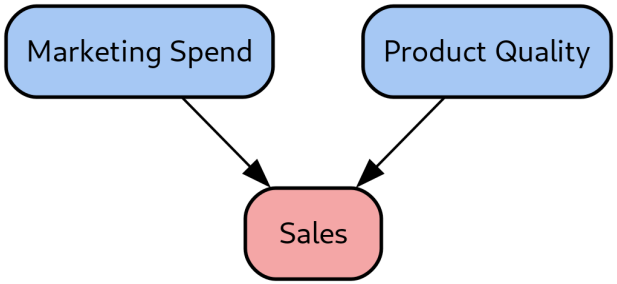


- **Example**
  - *Lifestyle* is a confounder that affects both *Weight* and *BloodPressure*
  - These outcomes may appear correlated due to shared cause



Inverted Fork

- An **inverted fork** occurs when two or more arrows converge on a common node
  - **Colliders** block associations unless the collider or its descendants are conditioned on
- **Conditioning on a collider** *opens a path*, inducing spurious correlations
  - This is the basis of selection bias



- **Example**
  - Sales influenced by multiple independent causes
  - *MarketingSpend* and *ProductQuality* both influence *Sales*
  - Conditioning on *Sales* can induce false dependence between *MarketingSpend* and *ProductQuality*